

Description

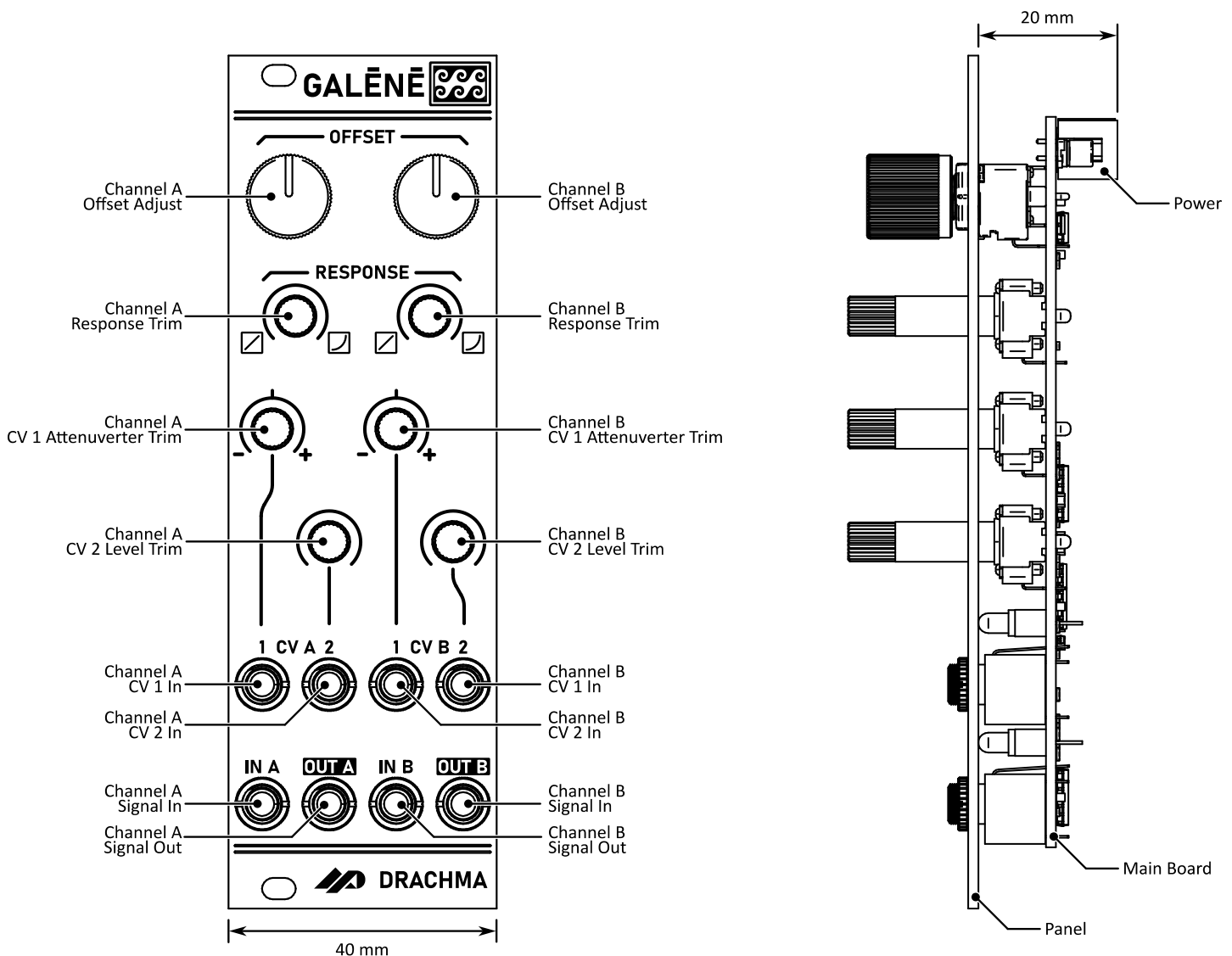
GALĒNĒ is a dual-channel, SSI2164-based VCA in 8hp. Both VCA channel inputs are DC-coupled and can process both audio and CV signals.

Each channel has 2 summed CV inputs with Level trim controls, Linear/Exponential Response trim, and Offset adjust from 0-5V. One CV input on each channel passes directly through an attenuverter before being summed to simplify sidechain patches.

Features

- SSI2164-based architecture
- 2 independent DC-coupled VCA channels
- 2 summed CV inputs per channel:
 - Level adjust trim per CV input
 - One attenuverter CV input per channel
- CV Offset adjust up to +5V
- Linear/Exponential Response adjust
- Indicator LEDs on both CV inputs and signal outputs
- Current Draw: +12V: 93 mA
-12V: 55 mA

Module Layout



Functionality

Signal Inputs/Outputs:

The signal inputs of Channel A and Channel B are DC coupled and can accept both CV or audio signals. Input signals can range from -10V and +10V. Larger signals may cause unwanted effects or distortion but will not damage the module.

Signal output level is indicated by the bicolor LED above the jack. A green color indicates positive output voltage, while red indicates negative output voltage.

CV Inputs:

Both channels have two CV inputs that are summed before going into the VCA gain network. One CV input on each channel passes through an attenuverter to adjust the CV signal polarity.

CV input levels can range from 0-10V: 0V will provide a gain of -100dB, 5V will provide unity gain, and 10V will provide +6dB gain. A negative summed CV signal has no further effect on reducing signal gain.

A red LED above the CV jacks indicates the summed CV level of both CV inputs for each channel. Negative CV values are not indicated by the LED.

CV Level Trim:

Each CV input can be independently trimmed using the Level trim knobs. The CV 2 trim knob acts as a simple attenuator, whereas the CV 1 trim knob of each channel controls the attenuverter and can invert the CV 1 signal.

CV Response Trim

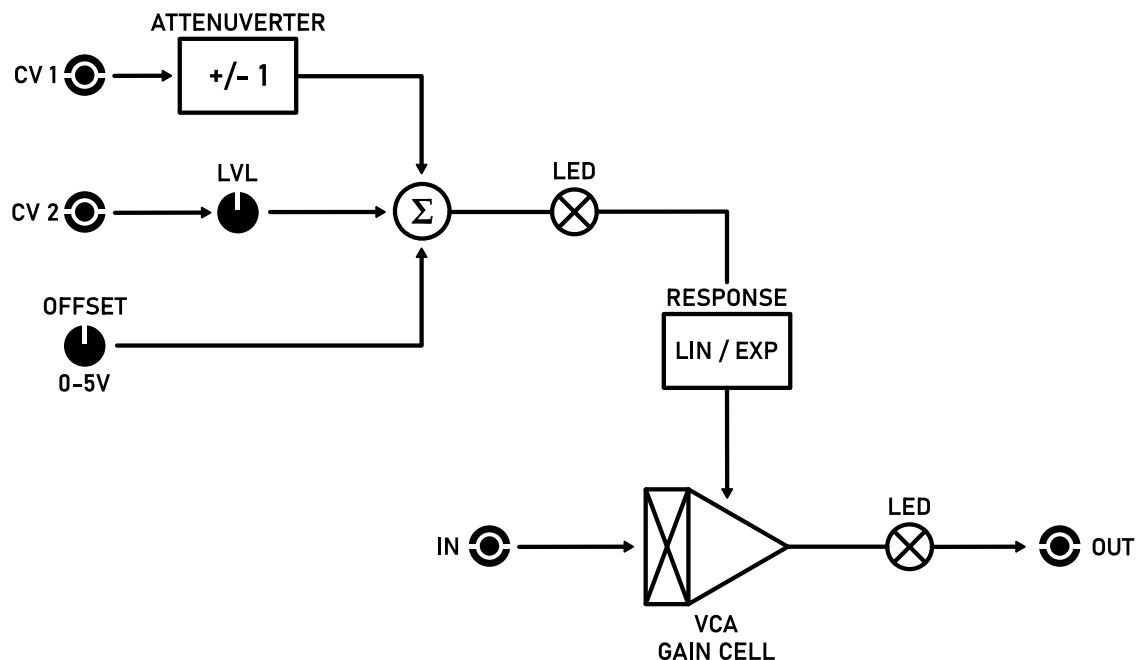
The response of the CV inputs for each channel can be controlled using the Response trim knobs. Turning the knob fully left will give a linear response to the CV input, while turning fully right will give an increasingly sharp exponential response. A sharp exponential response can saturate or distort the input signal.

CV Offset Adjust:

A voltage offset can be added to the CV inputs using the Offset adjust knobs. The resulting CV signal will be the sum of both CV inputs and the Offset voltage. The CV signal can be offset by up to +5V.

The Offset adjust used in conjunction with the CV 1 input attenuverter can greatly simplify sidechain or ducking patches.

Block Diagram (Single Channel)



Bill of Materials

Value	Qty.	Name on Board	Description	Footprint
220Ω	6	R23, R25, R27, R29, R34, R35	220Ω 1% 0.125W Resistor	0805
1kΩ	6	R6, R7, R33, R41, R47, R50	1kΩ 1% 0.125W Resistor	0805
10kΩ	4	R38, R44, R46, R48	10kΩ 1% 0.125W Resistor	0805
20kΩ	4	R21, R31, R51, R52	20kΩ 1% 0.125W Resistor	0805
27kΩ	2	R3, R32	27kΩ 1% 0.125W Resistor	0805
100kΩ	30	R1, R2, R4, R5, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R22, R24, R26, R28, R30, R36, R37, R39, R40, R42, R43, R45, R49	100kΩ 1% 0.125W Resistor	0805
100pF	4	C9, C10, C21, C22	100pF 50V C0G MLCC	0805
1.2nF	4	C11, C13, C14, C15	1.2nF 50V C0G MLCC	0805
100nF	12	C3, C4, C5, C6, C7, C8, C12, C16, C17, C18, C19, C20	100nF 50V X7R MLCC	0805
10uF	2	C1, C2	10uF 25V Elec. Capacitor	4x5.7mm SMD
1N4148 Diode	2	D5, D6	1N4148 Gen. Purp. Diode	SOT-23
LM4040	1	U1	LM4040 5V Reference	SOT-323
TL072	1	U2	TL072 Dual Op Amp	SOIC-8
TL074	4	U3, U4, U6, U7	TL074 Quad Op Amp	SOIC-14
SSI2164	1	U5	SSI2164 Quad VCA	SOIC-16
B10k Pot Lin	2	RV1, RV2	10k Lin. Pot, Round Shaft	RD901F
B10k Pot Lin	4	RV3, RV4, RV7, RV8	10k Lin. Trim Pot	RK09K
B100k Pot Lin	2	RV5, RV6	100k Lin. Pot	RK09K
3.5mm Jack	8	J1, J2, J3, J4, J5, J6, J7, J8	3.5mm Mono Jack	PJ-3001F
3mm LED	2	D1, D2	3mm LED Red	LED 3mm
3mm LED	2	D3, D4	3mm LED Bicolor Red/Green	LED 3mm
16-Pin Header	1	J9	IDC Header 16-Pin	IDC 16-Pin 2.54mm

